

SAVINDU SURANJITH THENNAKoon

savindus.23@cse.mrt.ac.lk | +94 723890494 | savindu.cse23.org | [Github](#) | [Linkedin](#)

SUMMARY

Motivated Computer Science & Engineering undergraduate at the University of Moratuwa with a strong foundation in algorithms, data structures, and system design. A fast learner and adaptable problem-solver who picks up new tools, languages, and frameworks quickly, with hands-on exposure to full-stack development, Mobile development, and artificial intelligence. Skilled in cross-functional teamwork, clear communication, and rapid prototyping, with a genuine curiosity for how systems work end-to-end. Eager to bring a results-driven, always-learning mindset to a Software Engineering Internship and contribute to impactful, real-world projects.

EDUCATION

University of Moratuwa Mar 2024 - Aug 2028 (Expected)
BSc (Hons) in Computer Science and Engineering, General Stream

Bandarawela Dharmapala College 2022
G.C.E Advanced Level - Physical Science Stream
• 3A passes | Z-score : 2.12 | District Rank : 16

TECHNICAL SKILLS

- Languages: C++, Javascript, TypeScript, Python, SQL
- Frontend: React, Vite, Tailwind CSS
- Backend: Node.js, RESTful API Design, JWT Authentication, FastAPI, RabbitMQ
- Databases: MySQL, PostgreSQL, Supabase, MongoDB, ChromaDB
- DevOps & Cloud: Docker, Git, GitHub Actions, Linux, Kubernetes
- AI/ML : LangChain, LangGraph, MediaPipe, TensorFlow.js, OpenCV, RAG Pipelines, Multi-Agent Systems

COMPETITIONS AND ACHIEVEMENTS

- Hackerrank orchestrate May (top 6%) - 81 out of 1400+ submissions
- Code Arena organized by IET | KDU - top 10
- Sliot Challenge 2026 - (top 50)
- DataStorm - Rotaract club of UOM - (top 30)

PROJECTS

Smart Attendance and Classroom Access Platform- (ongoing)



Tech Stack: React Native, React, Next.js, Supabase, Tailwind CSS, Face Recognition, GPS Geofencing

- Platform Architecture: Built an enterprise-grade academic attendance platform with a microservices architecture.
- Edge AI & Anti-Spoofing: Designed an on-device verification pipeline leveraging MiniFASNet, facial landmark tracking, and 192D embeddings to detect presentation attacks in real time.
- Fraud Prevention: Combined biometric checks with GPS geofencing and randomized verification windows to eliminate proxy attendance.
- Privacy-by-Design: Computed vector embeddings directly on-device, preventing raw facial imagery or biometric templates from leaving the client.
- Concurrency & Scalability: Offloaded vector matching to distributed RabbitMQ message queues for asynchronous processing during high-volume check-in bursts.

FitLock (AI App-Locker) - Live on google play (Feb 2026)



Tech Stack: React, Vite, Capacitor, MediaPipe Pose, On-Device ML

- Developed FitLock, a cross-platform digital wellbeing application that locks target apps behind physical exercise requirements (squats, push-ups, jumping jacks).
- On-Device Computer Vision: Integrated MediaPipe Pose via HTML5 Canvas for client-side, zero-latency repetition counting and joint angle calculation without streaming or storing user video.
- Monetization & Privacy Compliance: Integrated Google AdMob alongside the Google User Messaging Platform (UMP) SDK to handle monetization with full GDPR consent compliance.
- Currently rebuilding this on React Native for better performance

CivicGuard - Disaster Response & Emergency Intelligence Platform (Sep 2026)



Technologies: TypeScript, Node.js, Express, Python, FastAPI, React 18, PostgreSQL, PostGIS, Supabase, Google Gemini 3.5 Flash-Lite, YOLOv8, Kong API Gateway, Docker Compose, Socket.IO

- Core Functionality & Architecture: Built a nationwide disaster response platform utilizing an event-driven microservices architecture (Node.js, FastAPI, Kong Gateway, React 18) and PostGIS to automate incident triage and coordinate 250 tactical field squads across 25 simulated administrative districts.
- Technical Depth / Engineering Challenge: Implemented a 5-signal hybrid verification engine by coupling Gemini 3.5 Flash-Lite multimodal reasoning with YOLOv8 spatial telemetry and PostGIS spatial clustering, evaluating citizen hazard reports under a 1.5s inference SLA and validating them against live hydrological sensor feeds.
- Concurrency & Systems: Prevented race-condition over-allocation across emergency shelters during evacuation surges by designing an atomic bed reservation engine using PostgreSQL transactional row-level locking.



AI Co-Founder

Technologies: Python, TypeScript, FastAPI, LangGraph, Next.js 14, Pydantic v2, PostgreSQL 15, Redis 7, ChromaDB, Docker Compose, Tailwind CSS

- Engineered an autonomous multi-agent venture advisory platform using FastAPI, LangGraph, and Next.js 14, orchestrating 5 encapsulated domain modules via a central, typed Pydantic state machine (VentureState) to deconstruct raw startup ideas into structured business model canvases, market analysis, and 90-day execution roadmaps.
- Eliminated LLM arithmetic hallucinations in financial and GTM planning by building a deterministic Python calculation engine that enforces hard mathematical invariants.
- Containerized the full-stack system across 4 services using Docker Compose with integrated healthchecks, and built an asynchronous Server-Sent Events (SSE) streaming pipeline with Human-in-the-Loop (HITL) pause-and-resume endpoints for real-time stage telemetry.

HackerRank Orchestrate (May- 2026)



Tech Stack: Python, LangGraph, Google GenAI (Gemma/Gemini), ChromaDB, HuggingFace all-MiniLM-L6-v2, Pydantic, Asyncio

- Deterministic State Machine: Built a multi-node LangGraph workflow orchestrating classification, safety checks, policy routing, and ticket resolution across HackerRank, Claude, and Visa domains.
- Local RAG Pipeline: Indexed multi-domain support corpora into an embedded ChromaDB using HuggingFace all-MiniLM-L6-v2 embeddings, ensuring grounded, hallucination-free retrieval without external web dependencies.
- Type-Safe Routing & Structured Output: Utilized Google Generative AI paired with Pydantic schema models (.with_structured_output) to enforce strict CSV classification and triage decision compliance.
- Resilience & Concurrency: Designed an asynchronous processing engine using asyncio.Semaphore and staggered scheduling alongside Tenacity exponential retry policies to handle high ticket volume and transient API rate limits.
- Ranked 81st out of 1400+ submissions (12000 + registrants)

CERTIFICATES

- Professional Agentic Architect Certification (ongoing) | Google cloud

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Leadership Roles

- Marketing Committee Lead | MazeX 1.0 - Micromouse competition (2026)
- MATLAB webinar series - IET | Co-chair (July 2026)

Societies

- IEEE RAS | Web and Marketing pillar member (Feb 2026 – Present)
- Moraspirit Marketing pillar member (March 2025 – Present)
- Operational pillar Member | IET On Campus (March 2026 – Present)

Sports

- Team Member | Taekwondo Team, University of Moratuwa (2025)

REFERENCES

- Prof. Dulani Meedeniya

Senior Lecturer Department of Computer Science and Engineering University of Moratuwa

Email: dulanim@cse.mrt.ac.lk

- Dr. Adeesha Wijayasiri

Senior Lecturer Department of Computer Science and Engineering University of Moratuwa

Email: adeeshaw@cse.mrt.ac.lk